Transition convex hulls can differ for planar similarities

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Task 1. Resolve MathDB 354470. Conjecture 6.2 of Leśniak, Snigireva, Strobin and Vince [1], p. 5425, proposes equality of the metric convex hulls of the lower and upper transition attractors when each map takes metrically convex sets onto metrically convex sets. We give a counterexample on the Euclidean plane satisfying (H1)–(H3).

Proposition 1. *Let $e = (1, 0)$ and $J(x, y) = (-y, x)$. For $0 \le t \le 1$ set*

$$r = r_t = \frac{1+t}{2}, \qquad \varepsilon = 1 - r,$$

and define the two-map IFS $F_t = \{f_t, g_t\}$ by

$$f_t(x) = rJ(x - \varepsilon e) + \varepsilon e,$$
$$g_t(x) = rJ^{-1}(x + \varepsilon e) - \varepsilon e.$$

Its lower transition attractor is $A_\bullet = \{0\}$. It has at least one upper transition attractor, and every upper transition attractor contains $(1, -1)$. Hence their metric convex hulls differ.

Proof. The maps are distinct invertible similarities for every parameter, including both endpoints. Their coefficients depend continuously on t, proving (H1), and their Lipschitz constants are $r < 1$ for $t < 1$, proving (H2). Their respective unique fixed points are εe and $-\varepsilon e$, both tending to zero, proving (H3). Thus the limiting fixed-point set is $Q = \{0\}$.

In the Euclidean metric, metric segments are ordinary line segments. Every invertible affine map u satisfies $u([x, y]) = [u(x), u(y)]$, so both maps send every metrically convex set onto a metrically convex set, as required in the conjecture.

At $t = 1$, we have $F_1 = \{J, J^{-1}\}$. The singleton $\{0\}$ is invariant and contains Q, and every set containing Q contains it. The lower-transition definition therefore gives $A_\bullet = \{0\}$.

We next verify that upper transition attractors exist. Since $J^{-1} = -J$, the translations in the two affine maps are

$$b_+ = \varepsilon(e - rJe), \qquad b_- = -\varepsilon(e + rJe).$$

Their norms are $\varepsilon\sqrt{1 + r^2} \le 2\varepsilon$. Consequently the closed Euclidean ball $B = \{x : \|x\| \le 2\}$ is invariant under both maps, because

$$\|f_t(x)\|,\ \|g_t(x)\| \le 2r + 2(1 - r) = 2 \qquad (x \in B).$$

For $t < 1$ the contraction theorem gives a nonempty compact attractor $A_t \subset B$. The space of nonempty compact subsets of the compact set B is compact in the Hausdorff metric. Starting, for example, with $t_n = 1 - 1/n$, $n \ge 1$, we may pass to a subsequence along which A_{t_n} converges. This gives an upper transition attractor as in Definition 4.3 of the source.

To distinguish every such limit from the lower transition set, consider the composition $P_t = f_t \circ g_t$. Direct calculation gives

$$P_t(x) = r^2 x + \varepsilon\big((1+r^2)e - 2rJe\big). \tag{1}$$

Its unique fixed point, for $t < 1$, is

$$p_t = \frac{(1+r^2)e - 2rJe}{1+r}, \qquad \lim_{t\uparrow 1} p_t = e - Je = (1,-1). \tag{2}$$

For any $a \in A_t$, invariance gives $P_t^n(a) \in A_t$ for every n. Since P_t is a contraction, these points converge to p_t; closedness gives $p_t \in A_t$.

Now let $A^\bullet = \lim_n A_{t_n}$ be any upper transition attractor. Hausdorff convergence and (2) imply

$$\operatorname{dist}((1,-1), A^\bullet) \le \|(1,-1) - p_{t_n}\| + h(A_{t_n}, A^\bullet) \longrightarrow 0.$$

As $A^\bullet$ is closed, $(1,-1) \in A^\bullet$. In the source's notation, $\operatorname{conv}_d S = \bigcup_{x,y\in S}[x,y]$. In particular it contains S, because $[x,x] = \{x\}$. Hence

$$\operatorname{conv}_d A_\bullet = \{0\}, \qquad (1,-1) \in \operatorname{conv}_d A^\bullet,$$

which disproves the conjectured equality. □

The same formulas extend to all $t \ge 0$. For $t > 1$, f_t expands distances from its fixed point εe by the factor $r > 1$, so a generic orbit is unbounded. Thus F_t is not contractive in any equivalent complete metric, and cannot have an attractor with an open basin. At $t = 1$, J^4 is the identity, also precluding contraction in any metric. The finite contractivity threshold required in Section 4 is exactly one.

The centers vary with the parameter. Thus the construction meets the general conjecture while lying outside the older semilinear fixed-center hypothesis. It also works if one replaces the source's union-of-segments hull by the ordinary convex hull or its closure.

Task 2. Determine the complete limiting set. Establish whether this family's attractors have a unique Hausdorff limit, and calculate that limit explicitly.

Task 3. Quantify the effect of moving fixed points. For opposite rotations, characterize which rates and directions of center motion produce nonzero limiting fixed points of compositions.

Task 4. Identify sufficient hypotheses for hull equality. Determine which restrictions on moving centers extend the known fixed-center results to more general similarity families.

Task 5. Compare metric convexity conventions. Study equality of transition hulls defined by one union of segments, iterated segment closure and closed metric convexity in non-Euclidean spaces, under suitable additional hypotheses.

References

[1] K. Leśniak, N. Snigireva, F. Strobin and A. Vince, *Transition phenomena for the attractor of an iterated function system*, Nonlinearity **35** (2022), 5396–5426. doi:10.1088/1361-6544/ac8af1.